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Original article

Prevalence and trends of hepatitis C virus among blood donors in Antananarivo, from 2003 to 2009

Prévalence de l'infection au virus de l'hépatite C chez les primodonneurs à Antananarivo, de 2003 à 2009

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Abstract

Purpose of the study. – Due to anemia and maternal, childhood and infancy diseases, blood transfusion is one of the most important medical cares given in sub-Saharan Africa, including Madagascar. World Health Organization encourages worldwide countries to maximize transfusion security. Until now, there was no data within a large population of blood donors. Therefore, the aim of this study was to assess the prevalence and time trends of hepatitis C virus (HCV) among all first-time blood donors coming into the National Centre of Transfusion Supply in Antananarivo.
Material and methods. – This retrospective study was conducted at the National Centre of Transfusion Supply in Antananarivo from 2003 to May 2009. We looked up to all recorded results of systematic screening of blood donor candidates and we took data about all first-time blood donors, including age and gender. Forty-seven thousand five hundred and ten of 47,636 first-time blood donors were retained, the others were excluded due to lack of some data.
Results. – The mean age of our donors was 33.3 years (35.8 for male, and 32.6 for female; $P \ll 0.05$). Eighty percent were male (38,225/47,510). HCV prevalence was 0.65% during the period of study and HCV positive donor candidates were older than HCV negative (mean age: 39.1 vs. 33.2; $P \ll 0.05$). It was in fact stable from 2003 to 2007, and then decreased. HCV prevalence was higher in women than in men (0.9 vs. 0.6; $P < 0.05$), and it increased by age ($P < 0.05$).
Conclusion. – Our study found a low prevalence of HCV among blood donors compared to many countries in Africa.
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Keywords: Hepatitis C; Blood donors; Madagascar; Epidemiology; Transfusion

Résumé

But de l'étude. – L'acte transfusionnel est l'un des moyens thérapeutiques les plus importants en Afrique subsaharienne, y compris Madagascar, à cause de l'anémie et des maladies materno-infantiles. L'Organisation mondiale de la santé (OMS) prône une politique de sécurisation des dons au niveau mondial. Jusqu'à maintenant, peu de données sur une population de taille élevée étaient disponibles à Madagascar. D'où l'objectif de cette étude d'étudier la prévalence dans le temps de l'infection au virus de l'hépatite C (VHC) chez les primodonneurs au niveau du Centre national de transfusion sanguine d'Antananarivo.
Matériel et méthodes. – Il s'agit d'une étude rétrospective menée sur une période allant de 2003 à mai 2009. Nous avons collecté les données biologiques mentionnées dans le cahier de registre du Centre national de transfusion sanguine d'Antananarivo. Nous avons retenu 47 510 sur les 47 636 premiers dons de cette période, les autres ont été exclus pour cause de données incomplètes.

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Résultats. – L'âge moyen des donneurs était de 33,3 ans (35,8 pour les hommes et 32,6 pour les femmes ; $p « 0,05$). Quatre-vingts pour cent des donneurs étaient de sexe masculin (38 225/47 510). La prévalence de l'infection à VHC était de 0,65 %, les sujets VHC positifs étaient plus âgés que les sujets VHC négatifs (moyenne d'âge de 38,1 contre 33,2 ; $p « 0,05$). En réalité, la prévalence était stable de 2003 à 2007, ensuite elle a diminué. Enfin, la prévalence de VHC est plus élevée chez les femmes (0,9 contre 0,6) et elle augmente avec l'âge ($p < 0,05$).
Conclusion. – Notre étude montre une faible prévalence de l'hépatite C par rapport aux autres pays africains.
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Mots clés : Hépatite C ; Premiers dons ; Madagascar ; Épidemiologie ; Transfusion

1. Introduction

In 21st century, anemia remains a major public health in developing countries, especially in sub-Saharan Africa where 32 to 42% of women have anemia, and about 25% of postpartum deaths are related to obstetrical hemorrhages [1]. Therefore, finding the safest potential blood donor is very important, especially in these countries. The members of European community still have nowadays a policy about blood safety [2].

As in many countries in Africa, most of blood donations come from replacement and familiar donors in Madagascar. Antananarivo is the capital of Madagascar and about two million people live there.

In contrast to wealthy countries where more than 97% of transfused blood comes from voluntary donors [3], since 1969, the databook of the National Centre of Transfusion Supply (NCTS) in Antananarivo recorded only 7417 volunteer, non-remunerated blood donors who gave blood regularly. These few people ensure about one quarter to one third of all donations at the NCTS at Antananarivo.

According to the World Health Organization (WHO) guidelines, hepatitis C virus (HCV) screening was first introduced in the panel of serology testing in 2003. Previous data showed that before it was routinely screened, HCV was the main source of transfusion-transmitted hepatitis in industrialized nations and then it has dramatically fallen [4–6]; but what about Madagascar? There are only a few available data. One study conducted in two villages of Madagascar found HCV prevalence ranging from 2.2 to 5.8% [7]. In 2008, HCV prevalence among general population living in Antananarivo was about 1.2% [8]. For blood donors, a study was conducted from 2001 to 2003 at the NCTS in Antananarivo by using HCV tri-dot, a fourth generation immunochromatographic testing and the authors found about 0.8% of HCV positive [9]. Unfortunately in this study, first-time donors were not separated from regular donors.

Therefore, the aim of this study was to assess the prevalence of HCV infection among all first-time blood donors coming to the NCTS in Antananarivo.

2. Materials and methods

This is a retrospective study, conducted at the NCTS located in Antananarivo, the capital of Madagascar, from 2003 to May 2009. The reason why we chose 2003 as the beginning of this study is that HCV screening with Flavicheck® reagent (Orchid Biomedical System, India) was introduced at this time into routine screening of transfusion-transmitted disease among blood donation in Madagascar. And we stopped our study on May 2009, because the NCTS in Antananarivo changed the rapid testing kit of HCV screening at this date.

We looked up to all recorded results of systematic serology screening of blood donor candidates, according to the national policy for blood donation in Madagascar, with serology for hepatitis B and C, human immunodeficiency virus (HIV) and syphilis. Data included also age and sex of each donor. We included all first-time blood donors, including new volunteer blood donors and replacement blood donors.

2.1. Blood donation candidates selection

The first step of blood donation consisted of interviewing each candidate according to a questionnaire, which aim is to exclude all candidates who have a potential risk for transmitting disease by their behavior and sexual attitudes. In the NCTS in Antananarivo, the donor screening and selection are performed by physicians. So, deferral donors candidates are those who are polytransfused, or who had a recent blood transfusion (during the last 2 years), those who got recent fever (during the last 15 days), including paludism, treated or not, those who had a history of icterus, tuberculosis, prolonged fever, dermatitis, adenopathy, sexually transmitted disease, or any cause of bleeding, any cardiovascular disease, bronchopneumopathy, allergy or recent travelling abroad. Finally, those who got piercing, tattoo, injection or dental care, or were homosexuals, or had multiple partners and other high-risk sexual behavior such as unprotected sexual intercourse with new partners, were excluded.

It is important to note here that in 2007, a team from the Belgium Red Cross came to the NCTS in Antananarivo and trained all workers about all donation steps (from drawing the questionnaire to serological testing). After that, the team at the NCTS in Antananarivo used a new questionnaire for donor interviews.

All people who did not know anything about HIV/AIDS infection got counseling by physicians.

A physical examination was performed, and at this step, deferred donor candidates are those who had systolic blood pressure higher than 180 mmHg, heart frequency higher than 120 bpm, weight less than 48 kg, and any clinical symptoms of dermatosis or any clinical symptoms of anemia (screening hemoglobin level by biochemical or another technique is not yet including in routine).

Nonetheless, some donors were temporarily excluded, such as women during their menstruation, a person who slept less than 4 hours, recent alcohol drinking (less than 72 hours), dental care

within the last 6 months, and if blood pressure was just taken after a physical effort.

Blood was collected then screened for blood-borne infectious agents.

For our study, a first-time blood donor is identified as a donor who donated for the first time.

2.2. Serological testing

Serum samples were screened for blood-borne infectious agents including hepatitis C for which screening was performed by using rapid testing (Flavicheck®, Orchid Biomedical System, India). The rapid testing used for HCV screening was the same throughout the period of our study. The test was done according to the manufacturer's guidelines. A double readout was performed, the first by the laboratory technician who realized the testing, and the second by the physician. There was no confirmatory strategy for HCV reactive sera in the routine screening. Therefore, all reactive donations were excluded.

2.3. Statistical analysis

Prevalence rates were calculated by using univariate analyses, and then bivariate analyses using Chi^2 were both performed by using Excel and Epi Info 2000 Ver. 3.5.1. We retained as significant result those who had P value less than 0.05.

2.4. Ethical issues

Due to the nature of the study, a retrospective review of blood donors, no informed consent of individuals included in the study was needed.

3. Results

During the period of study, 47,636 first-time donations were registered in the databook. Data for 47,510 individuals were finally retained; the others were excluded due to incomplete information such as gender or age.

The mean age of donors was 33.3 ± 10.4 years (35.8 ± 10.9 for male, and 32.6 ± 10.1 for female; $P < 0.05$); 67.22% of them were aged of 20 to 40 years; 80.46% were male (38,225/47,510) (Table 1).

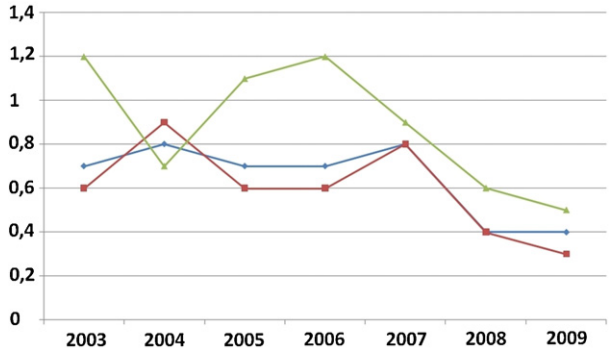


Fig. 1. Hepatitis C virus prevalence from 2003 to 2009: (■): male; (▲): female; (♦): total.

HCV prevalence during the whole period of study was 0.65% (95 IC% 0.6–0.7; 309/47,510). HCV positive donor candidates were older than HCV negative (mean age: 39.1 ± 11.5 vs. 33.2 ± 10.3), but by analyzing it, we found that HCV prevalence was quite stable from 2003 to 2007 (0.69 to 0.80%), then fell since 2007 (0.44% in 2008 and 0.37% in 2009; $P < 0.05$) (Fig. 1). We also found that there was a significant predominance of prevalence among women compared to men in HCV positive donor candidates (0.9% vs. 0.6%; $P < 0.05$). There was also a significant relationship between age and HCV seropositivity ($P < 0.05$): in general it increased with age both in men and women.

4. Discussion

Our study may be limited by many facts. First of all, it was a retrospective study and all biases of a retrospective study may interfere with our results. Secondly, performing blood testing in Madagascar by rapid testing could underestimate the real prevalence of this disease in our country. Immunologic assays are not very reliable for HCV diagnosis: they could underestimate it [10]. On the other hand, rapid test can overestimate it [11]. Moreover, all rapid tests are not reliable to diagnose HCV infection due to their low sensitivity and their low specificity compared to EIA techniques [11,12]. That is why industrialized countries use molecular techniques to diagnose HCV among donors [13].

All policies about blood transfusion aim to ensure and maximize safety in blood transfusion and some criteria must be ful-

Table 1
General prevalence of hepatitis C virus (HCV) seroprevalence by age group.

	Male			Female			Total		
Age group	HCV negative	HCV positive	%	HCV negative	HCV positive	%	HCV negative	HCV positive	%
< 20	2973	12	0.40	669	3	0.45	3657	15	0.41
20–25	8642	28	0.32	1442	9	0.62	10,121	37	0.37
25–30	7277	30	0.41	1236	7	0.56	8550	37	0.43
30–35	5928	20	0.34	1354	6	0.44	7308	26	0.36
35–40	4641	31	0.67	1271	14	1.09	5957	45	0.76
40–45	3523	34	0.97	1216	11	0.90	4784	45	0.94
45–50	2556	30	1.17	995	14	1.39	3595	44	1.22
> 50	2456	44	1.79	1022	16	1.54	3538	60	1.70
Total	37,996	229	0.60	9205	80	0.86	47,510	309	0.65

filled by the donor before sampling their blood and screening it. Thus, blood collection and screening are centralized at regional centres that are independent of individual hospitals. These centres recruit voluntary blood donors, screen and group blood, process it into components and have distribution networks to provide regular supplies to hospitals in their region [14]. Answering a questionnaire is the first step of securing blood and many countries have their criteria [15]. Then, all blood collected from donors passed through a serology screening. The decrease of HCV prevalence found in our donors can be explained by the fact that in 2007, some members of Red Cross from Belgium came into the NCTS in Antananarivo and trained all physicians working there about all donors screening steps. After that, physicians were more aware about all data collected by filling the questionnaire. Therefore, just by doing correctly the first step of screening and by being more aware about all data obtained in questions, the risk of transfusion-transmitted infections can be divided by two.

It seemed that Madagascar had a low HCV prevalence compared to many countries in Africa. A study conducted in seven countries found HCV seroprevalence varying from 1.42% in Niger to 6.98% in Ivory Coast, and first-time blood donors represent from 30 to 75.8% [14]. A prospective study in Nigeria showed an HCV seroprevalence of 3.1% among 1410 healthy blood donors 18 to 64 years old (mean: 32.58 ± 10.24 years) [16]. In Cameroon, 26.2% of all donors had at least one transfusion-transmitted infection and 4.8% of first-time blood donors were HCV positive [17]. In Mali, among 25,543 donations during 2007, 3.3% were positive for HCV [18]. In Egypt, 24.8% of blood donors were HCV positive, ranging from 6.0 to 28.8% [19,20]. Finally, in Iran, seroprevalence of HCV is about 0.13% [21].

The NHANES III study showed that 1.8% of healthier blood donor had HCV, varying from 1.5 to 3.2% depending of the race [22]. In Brazil, 0.97% of female, and 0.38% of male blood donor were HCV positive [23]. And in Mexico, among 61,553 first-time blood donors, the seroprevalence of HCV was 0.84% [24]. In Nepal, 0.64% of 21,716 blood donors were HCV positive and in Mongolia, the prevalence among blood donors was 9.6% [25,26]. In UK, only 0.53% of 287,332 donors showed a serum reactivity during the screening and among them, only 5% of reactive samples were confirmed true positive [27].

As seen in a majority of data collected about blood donors, male predominance was found [14,21,25,28]. Our study showed that most donors are young males; it is probably due to the fact that in Madagascar, blood transfusions are usually done in case of emergency; that is also the case of finding replacement donors. It is usual in Madagascar to ask at first for young people, especially men, within their own family to do this.

Interestingly, our study showed that HCV prevalence increased by age and women are more exposed. As in many countries in Africa, healthcare is very expensive in Madagascar, which is why Madagascans are used to come into hospitals only if there is no alternative solution for their disease. Nevertheless, that is not the case of pregnancy and birth. Women usually go to healthcare centres during pregnancy where they get care in order to avoid stillborn children. As HCV is essentially transmitted by contaminated materials, we hypothesize that this is the main source of women contamination here.

On the one hand, each country may have its own strategy. In Madagascar, the current national health policy about blood transfusion does not require repeating the screening to confirm a positive result. All positive samples are not validated. The second test is only performed in order to confirm the result which will be given to the donor. It may be interesting to do that for the final result because in Ghana for example, only 23 out of 154 positive samples at the first screening were confirmed as positive [29]. One the other hand, there are problems that blood banks of developing countries are facing: to be sure that all negative samples are truly negative, and the same way with positive samples.

Some strategies that aim at avoiding retesting all positive sampling are usually used. With Elisa techniques, analyzing the S/CO value could permit it [30]. But if Elisa is well known as having high sensitivity to detect antibodies, it is too expensive for most of developing countries that is why they usually use rapid testing to perform all analyses. Unfortunately, these rapid testings are not reliable due to their low sensitivity, compared to Elisa [11].

Besides, finding negative result is also not easy because of the window period. Many techniques could not find asymptomatic infected people during it. Therefore, new technologies based on molecular biology were developed [31]. Moreover, these last techniques, such as PCR, or recombinant immunoblot assay (RIBA) could also permit to avoid false-negatives due to the impossibility of Elisa or other techniques to detect a very low viraemia in order to make the diagnosis of HCV infection [32].

But the main obstacle is their cost. A possible alternative strategy is to test samples in a sequential order [33]: laboratories can either work with the best techniques and save money. One way to reduce the costs of all mandatory tests in blood screening is to pool sera and then to do molecular testing in the pool, such as PCR; but it needs standardization due to loss of sensitivity if there are too many samples within the pool [34].

However, none of the new technologies that are available nowadays and which are used in diagnosis of HCV infection could distinguish acute from chronic infection [35].

Unfortunately, we did not separate familiar and replacement donors from volunteer donor who came to the NCTS for the first time, it would be interesting to compare the prevalence among them.

5. Conclusion

We found a low prevalence of HCV positive donors and it highlighted the importance of questionnaire in screening donor candidates. We found also more HCV positive women, and the older the donor is, the higher the risk of HCV positivity is.

Disclosure of interest

The authors declare that they have no conflicts of interest concerning this article.

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